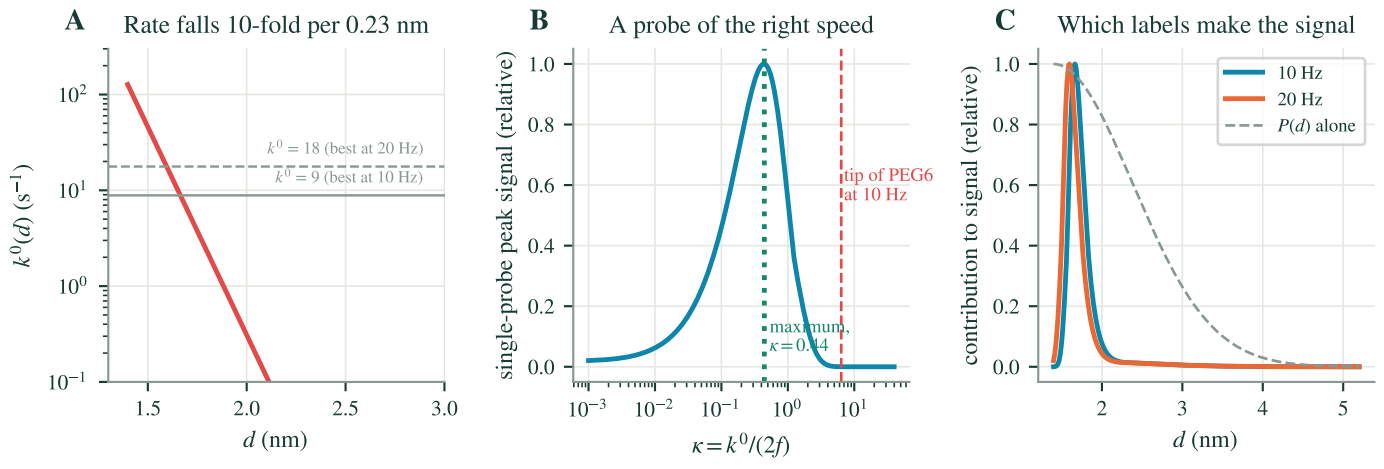
A
Rate falls 10-fold per 0.23 nm
$k^0 = 18$ (best at 20 Hz)
$k^0 = 9$ (best at 10 Hz)
$k^0(d)$ (s^{-1})
10^2
10^1
10^0
10^{-1}
1.5
2.0
2.5
3.0
d (nm)
B
A probe of the right speed
single-probe peak signal (relative)
1.0
0.8
0.6
0.4
0.2
0.0
tip of PEG6
at 10 Hz
maximum,
$\kappa = 0.44$
10^{-3}
10^{-2}
10^{-1}
10^0
10^1
$\kappa = k^0/(2f)$
C
Which labels make the signal
contribution to signal (relative)
1.0
0.8
0.6
0.4
0.2
0.0
10 Hz
20 Hz
$P(d)$ alone
2
3
4
5
d (nm)